
Outcome-Weighted Risk-Aware Supervised Learning for Chinese Standard Mahjong

Improving offense-defense trade-offs under reproducible paired
evaluation

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Baseline architecture: a supervised policy



Structured visible state

Hand, remaining tiles, four rivers and melds, wall progress, shanten, and fan-route cues.

Directly executable policy

One forward pass selects a legal action from 235 slots.

Evidence: IJCAI 2020 champion-Bot dataset, project architecture report, and competition record.

Why naive imitation still deals in

Observed failure

High-cost deal-ins erased gains from otherwise effective offense.

Learning mismatch

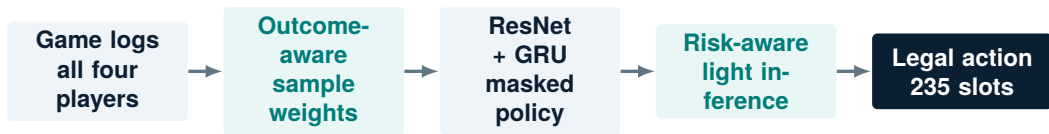
Cross-entropy treated every logged action as equally valuable, despite different long-term returns.

The question that drives the redesign

Not only **what** to imitate, but **what deserves more weight**.

Evidence: Mahjong Competition Rules setting and project action/state design.

Overall framework and three central innovations



1. Outcome weighting

Down-weight low-return and deal-in trajectories.

2. Temporal risk

Track each opponent and the true discard order.

3. Paired evidence

Control walls and seats before comparison.

Evidence: Project implementation: `preprocess.py`, `model.py`, `evaluate.py`.

Outcome weighting reduces low-return influence

Per-player terminal return

$$r_p = \tanh\left(\frac{s_p}{32}\right)$$

Policy weight

$$w_p = w_{\text{fan}} \left(0.20 + \frac{0.80}{1 + \exp(-3r_p)} \right) \times \begin{cases} 0.20, & p \text{ dealt in} \\ 1, & \text{otherwise} \end{cases}$$

Evidence: Exact weighting rule in `src/SL/preprocess.py`.

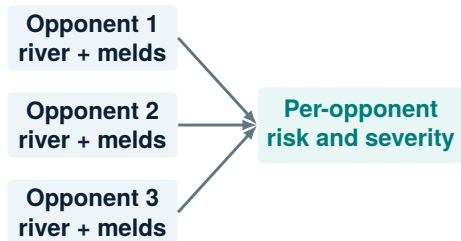
Preserve

Useful defensive choices from non-winners.

Suppress

Low-return trajectories and terminal deal-ins.

Risk is opponent-specific and time-sensitive



True table-wide order

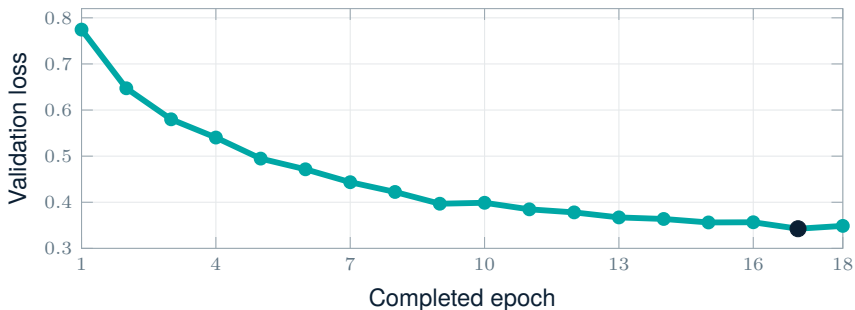
A GRU retains recency and post-meld evidence.

Light inference

Risk only reranks close discard candidates.

Evidence: Project implementation: per-opponent danger features, risk heads, and recency decay.

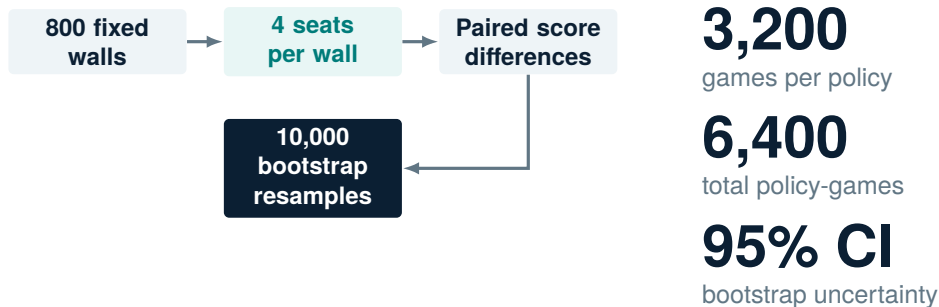
Supervised training loss curve



Validation loss: 0.77 → 0.34 saturation near epoch 17.

Evidence: Project supervised-training log.

Paired evaluation controls wall and seat variance



Primary statistic: same wall, controlled seat, frozen opponents,

$$\Delta_i = \text{SCORE}_{\text{final},i} - \text{SCORE}_{\text{baseline},i}.$$

Evidence: Final paired-evaluation per-game CSV, summary, bootstrap output, and manifest.

Outcome weighting improved both score and risk

+1.039

Avg. score (0.49 to 1.53)

+1.19 pp

Win rate (24.53% to 25.72%)

-3.19 pp

Deal-in rate (18.13% to 14.94%)

-0.059

Avg. rank (2.500 to 2.441)



Paired average-score improvement

Win rate rose while deal-in rate fell; the paired CI stayed above zero.

Evidence: Final paired evaluation, 800 walls $\times$ 4 seats per policy.

Ablation attributes the gain to the shared policy

Setting	Avg. score	Deal-in
Previous policy	0.934	17.95%
Outcome-weighted policy	1.772	15.70%
+ action-value inference	1.750	15.60%
+ all auxiliary controls	1.796	15.50%

2,000 games per setting; point estimates shown.

Robust contributor

Outcome-weighted training changed the shared policy and produced the consistent gain.

Negative result

Action-value, expected-loss, and fan-route controls lacked reliable incremental evidence.

Evidence: Six-setting ablation. Incremental auxiliary-control comparisons have CIs crossing zero.

Competition-tested, with a narrow claim

9th / 16

Competition final

2.37 ms

Mean decision time

Supported conclusion

The evidence supports a **practical, low-cost improvement** of a supervised-learning Bot.

Evaluation boundary. No matched-data state-of-the-art comparison is claimed.

Evidence: Competition record and final paired-evaluation manifest.

Three takeaways

1. Outcome-weighted imitation

Reduce low-return and deal-in noise.

2. Opponent-specific temporal risk

Use opponent identity and discard order to recognize danger.

3. Matched-wall paired evidence

Control wall and seat variance before policy comparison.

Next

Richer data and augmentation · adaptive aggression · self-play reinforcement learning

Evidence: Project conclusions; future work follows directly from the ablation limitations.

References

- [1] K. He, X. Zhang, S. Ren, and J. Sun. “Deep Residual Learning for Image Recognition.” *CVPR*, 2016.
- [2] J. Li et al. “Suphx: Mastering Mahjong with Deep Reinforcement Learning.” arXiv:2003.13590, 2020.
- [3] W. Li et al. “Tjong: A Transformer-based Mahjong AI via Hierarchical Decision-making and Fan Backward.” *CAAI Transactions on Intelligence Technology*, 2024.
- [4] D. Silver et al. “Mastering the Game of Go with Deep Neural Networks and Tree Search.” *Nature*, 529(7587):484–489, 2016.

Backup: final inference configuration

Enabled

- Outcome-weighted policy checkpoint
- Per-opponent risk head
- Light post-processing
- Learned tenpai weight = 1.0
- Legal-action firewall

Disabled after ablation

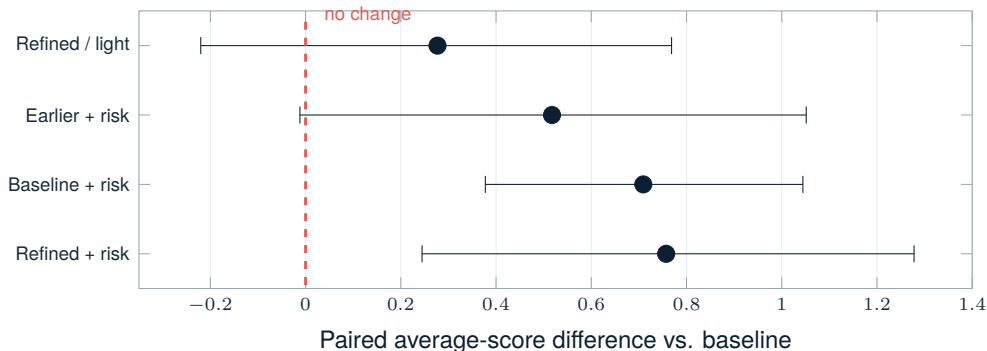
- Action-value inference
- Expected-loss head control
- Fan-route inference bonus
- Auxiliary discard-rank control

Checkpoint SHA-256

fa14dc7a045b9139fbc306668879d4073401a8f5251163e92d92920e7b7ef7dd

Evidence: Final evaluation manifest and the submitted inference configuration.

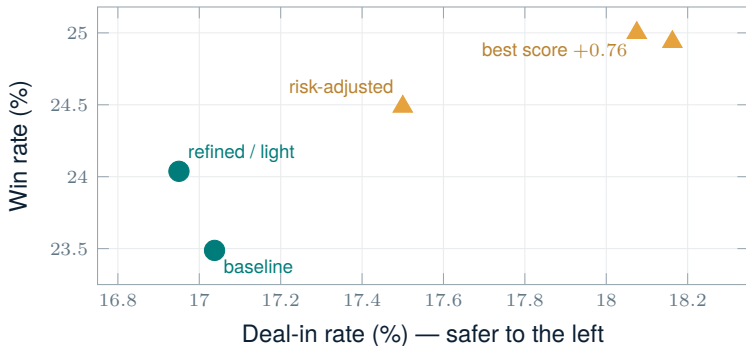
Backup: paired tests separated robust gains from noise



Only two risk-adjusted candidates cleared zero; the light refinement did not.

Evidence: Development-stage paired bootstrap results, 2,000 fixed walls $\times$ 4 seats per candidate. These are not the final headline-policy results.

Backup: higher development score did not always mean safer play



Development lesson

- Light refinement moved left and up
- Risk adjustment raised wins and score
- But deal-in rate also rose

Evidence: Development-stage candidate summary and per-game CSV, 8,000 games per candidate.

Backup: detailed final metrics

Metric	Old policy	Final policy	Change
Average score	0.4941	1.5331	+1.0391
Average rank (lower is better)	2.5003	2.4411	-0.0592
Win rate	24.53%	25.72%	+1.19 pp
Self-draw rate	7.13%	7.53%	+0.40 pp
Deal-in rate	18.13%	14.94%	-3.19 pp
Average deal-in fan	12.45	12.41	-0.04
Big deal-in rate	0.094%	0.094%	0.00 pp
Mean decision time	2.40 ms	2.37 ms	-0.03 ms
Invalid/timeout rate	0.875%	0.781%	-0.094 pp

Evidence: Final paired-evaluation summary CSV.

Backup: reproducibility package and remaining evidence

Already archived

- Fixed seed and 800-wall protocol
- Per-game CSV and summary CSV
- Paired bootstrap JSON
- Checkpoint hashes and software versions
- Exact final inference switches

Still needed for publication

- Official leaderboard URL or screenshot
- Commit hash and public release snapshot
- CI-backed component ablations
- Risk-calibration analysis
- Two traceable decision case studies
- KL-regularized self-play RL fine-tuning

Evidence: Project data-requirements audit.